

## LESSON 1 Matrices: Rows and Columns, Product and Factorization

MATH4570 is a second course in Linear Algebra, with a focus on **methods used in data science**. A basic goal of Linear Algebra is to solve the **matrix equation**  $Ax = b$ . In this lesson, we will deep dive into **matrix equations** and **matrices**, which are of fundamental importance in data analysis and machine learning.

### 1.1 Row Picture and Column Picture

We start with a small linear system/matrix equation and solve it by **Gaussian elimination**:

$$\begin{array}{ccc} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} & \Leftrightarrow & \begin{bmatrix} 1 & 1 & | & 5 \\ 2 & -1 & | & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & 2 \\ 0 & 1 & | & 3 \end{bmatrix} \quad \begin{array}{l} x_1 = 2 \\ x_2 = 3 \end{array} \\ A & x & b \qquad \qquad \qquad [A|b] \quad \text{rref}[A|b] \end{array}$$

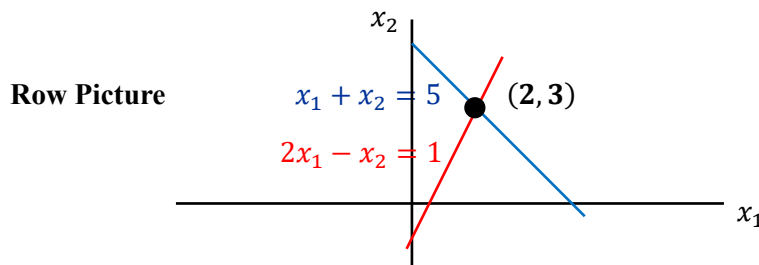
There are two ways to do the **matrix product**  $Ax$ , resulting in two pictures.

- **Rows (of  $A$ ) by columns (of  $b$ )**

$$Ax = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} (1,1) \cdot (x_1, x_2) \\ (2,-1) \cdot (x_1, x_2) \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ 2x_1 - x_2 \end{bmatrix} \quad \begin{array}{l} x_1 + x_2 = 5 \\ 2x_1 - x_2 = 1 \end{array}$$

$2 \times 2 \quad 2 \times 1 \qquad \qquad \qquad 2 \times 1$

Each **row** of  $Ax = b$  corresponds to a linear equation, which is a line in the  $xy$ -plane. The lines intersect at the **solution**  $(x_1, x_2) = (2, 3)$ ; this is the **row picture**.



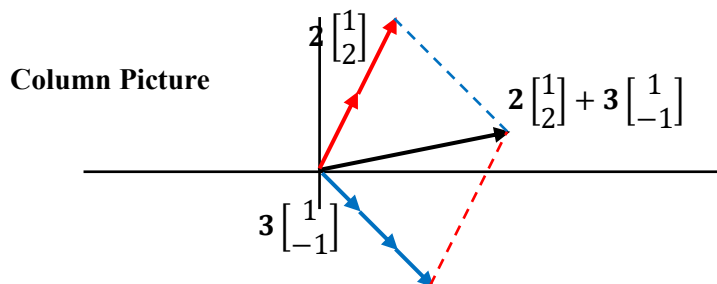
- **Columns (of  $A$ ) by rows (of  $b$ )**

There is another way of looking at  $Ax$ , by the **columns** of  $A$ :

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ 2x_1 - x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 2x_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ -x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} x_2$$

$Ax$  is a **linear combination** of the columns of  $A$ . In our example,  $(5,1)$  is a linear combination of  $(1,2)$  and  $(1,-1)$ , this is the **column picture**.

$$\begin{bmatrix} 5 \\ 1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



Let's look at a larger system:  $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & | & 2 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$ .

The system is consistent: in the last row  $0 = 0$ . The last column does not have a pivot and corresponds to a **free variable**  $x_3 = t$ , and we solve the **pivot variables** in terms of  $t$ :

$$\begin{aligned} x_1 + x_3 = 2 &\Rightarrow x_1 = 2 - x_3 = 2 - t \\ x_2 + x_3 = 1 &\Rightarrow x_2 = 1 - x_3 = 1 - t \end{aligned} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 - t \\ 1 - t \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Again, there are two ways of looking at  $Ax = b$ , by rows or columns of  $A$ :

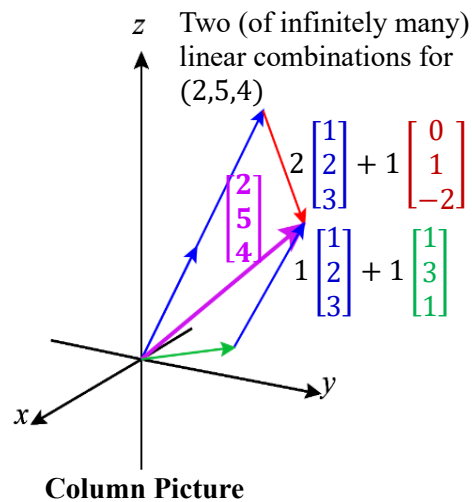
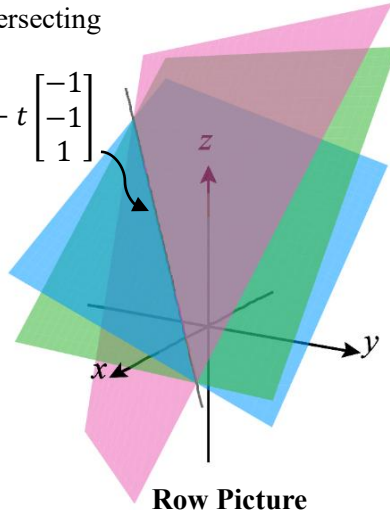
$$\begin{array}{ccc} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} & \begin{array}{l} x_1 + x_3 = 2 \\ 2x_1 + x_2 + 3x_3 = 5 \\ 3x_1 - 2x_2 + x_3 = 4 \end{array} & x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix} \\ A & x & b \end{array} \quad \begin{array}{c} \text{Rows} \\ \text{Columns} \end{array}$$

In the row picture, each equation is a plane, there are 3 planes **intersecting** at the solution space, which is a **line**:  $x = (2, 1, 0) + t(-1, -1, 1)$ . In the column picture,  $b = (2, 5, 4)$  is a **linear combination** of the columns of  $A$ , in infinitely many ways, two of which are:

$$\begin{aligned} t = 0: \quad (2, 5, 4) &= 2(1, 2, 3) + 1(0, 1, -2) + 0(1, 3, 1) \\ t = 1: \quad (2, 5, 4) &= 1(1, 2, 3) + 0(0, 1, -2) + 1(1, 3, 1) \end{aligned}$$

Planes intersecting at a line

$$x = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$



- **Rows:** each row of  $Ax = b$  is a **linear equation**, whose graph is a hyperplane, and there is a solution if and only if the hyperplanes **intersect**.
- **Columns:**  $Ax$  is a linear combination of the **columns** of  $A$ , and  $Ax = b$  has a solution if and only if  $b$  is a **linear combination** of the columns of  $A$ .

## 1.2 Column Space

Both the row and column pictures are important. In the first course we emphasize the row picture, which is about the **linear equations**. As the second course unfolds, we will see that the column picture becomes more prominent. The column picture is about **vectors**:  $b$  and the columns of  $A$ . Modern linear algebra is based on **vectors** and their **linear combinations**, which form a **vector space**. Recall that the **column space** of  $A$ , denoted  $C(A)$ , is the **span** (or the set of all linear combinations) of the columns of  $A$ . For an  $m \times n$  matrix  $A$ , the columns are vectors in  $\mathbb{R}^m$ , and  $C(A)$  is a subspace of  $\mathbb{R}^m$ . The **independent** columns of  $A$ , as identified by the pivots in  $\text{rref}(A)$ , form a **basis** of  $C(A)$ .

For our  $3 \times 3$  matrix  $A$ , there are pivots only in the first two columns of  $\text{rref}(A)$ ,  $C(A)$  is not  $\mathbb{R}^3$  but a 2-dimensional plane/subspace of  $\mathbb{R}^3$  spanned by the first two columns  $(1,2,3)$ ,  $(0,1,-2)$ , which form a **basis** of  $C(A)$ . The **column rank** of  $A$  is  $\dim(C(A)) = 2$ . Since  $C(A)$  is not the whole  $\mathbb{R}^3$ , there is  $b$  such that  $Ax = b$  has no solution  $x$ , for example, if  $b = (2,5,5)$ , we get  $0 = 1$  in  $\text{rref}[A|b]$ , and there is no solution:

$$\left[ \begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 5 \\ 3 & -2 & 1 & 5 \end{array} \right] \rightarrow \left[ \begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

- $Ax = b$  has a solution if and only if  $b$  is in the column space of  $A$ .

The row picture of the inconsistent system is 3 planes without a common intersection. In the column picture, the vector  $(2,5,5)$  does not lie on the  $C(A)$  plane.

### 1.3 Matrix Product and Factorization

Vectors  $u$  and  $v$  in  $\mathbb{R}^n$  may be viewed as matrices, and we may form the following products.

- **Inner product/dot product:**  $u \cdot v = u^T v$ , row times column, we get a scalar/number. (Inner product is more general than dot product, but we focus on dot product for now.)
- **Outer product:**  $uv^T$ , column times row, we get a matrix.

| Outer Product                               |                                           |                                                                       | Inner Product                             |                                             |                                     |
|---------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------|---------------------------------------------|-------------------------------------|
| $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ | $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ | $= \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$ | $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ | $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ | $= \begin{bmatrix} 6 \end{bmatrix}$ |
| $u$                                         | $v^T$                                     | $A = uv^T$                                                            | $u^T$                                     | $v$                                         | $u^T v$                             |
| $3 \times 1$                                | $1 \times 3$                              | $3 \times 3$                                                          | $1 \times 3$                              | $3 \times 1$                                | $1 \times 1$                        |

The outer product  $A$  is a  $3 \times 3$  matrix. The **column rank** of  $A$  is 1, and  $C(A)$  is one-dimensional, with basis  $\{(1,1,1)\}$ . The **row space**  $C(A^T)$ , is the column space of  $A^T$  and is one-dimensional with basis  $\{(1,2,3)\}$ . Observe that  $A$  is a product of  $C$  containing the basis of  $C(A)$  and  $R$  containing the basis of  $C(A^T)$ , this is **CR factorization**:

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$A = C R$$

Matrix product and factorization are the two sides of a coin: we multiply  $CR$  to get  $A$  or we factor  $A$  into  $CR$ . It is easier to multiply than to factor, but **matrix factorization** is important in data science. We put our data in a matrix  $A$  and factorize  $A$  in different ways to uncover information in  $A$ . The CR factorization may be viewed as a generalization of the outer product: we can do CR factorization for any matrix, not just “rank 1” matrices.

**CR factorization:** Any matrix  $A$  may be written as  $CR$ , where  $R = \text{rref}(A)$  without the zero rows and  $C$  is the independent columns of  $A$  corresponding to the pivots of  $R$ .

Before we give an example, recall that

$$A \begin{bmatrix} | & & | \\ c^{(1)} & \dots & c^{(m)} \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ Ac^{(1)} & \dots & Ac^{(m)} \\ | & & | \end{bmatrix}$$

Then, we may see CR factorization

$$\begin{array}{ccc} \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{2} & \mathbf{1} & \mathbf{3} \\ \mathbf{3} & \mathbf{-2} & \mathbf{1} \end{bmatrix} & = & \begin{bmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{2} & \mathbf{1} \\ \mathbf{3} & \mathbf{-2} \end{bmatrix} \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} \end{bmatrix} \\ 3 \times 3 & & 3 \times 2 \quad 2 \times 3 \\ A & = & C \quad R \end{array}$$

in terms of the columns of  $C$  spanning the columns of  $A$ :

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The row space of  $A$  is the same as the row space of  $\text{rref}(A)$ , spanned by the rows in  $R$ ; row operations take linear combinations of rows or swap rows and do not change the row space. The row space of  $A$  has dimension 2, this is the **row rank** of  $A$ . The number of columns in  $C$  has to be the same as number of rows in  $R$  for the product  $CR$ , so there are as many independent columns as there are independent rows, proving that the column rank and row rank are equal. In Math 2331, the rank of  $A$  is the number of pivots in  $\text{rref}(A)$ , now we see that the rank is the number of independent columns *as well as* independent rows. Moreover, each “non-pivot” column of  $R$  informs us how the corresponding column in  $A$  depends on the independent columns:  $\begin{bmatrix} \mathbf{1} \\ \mathbf{1} \end{bmatrix}$  in column 3 tells us that  $v^{(3)} = \mathbf{1}v^{(1)} + \mathbf{1}v^{(2)}$ .

#### 1.4 Four Fundamental Subspaces

We conclude the lesson by recalling the **four fundamental subspaces** associated with an  $m \times n$  matrix  $A$ : **column space**  $C(A)$ , **row space**  $C(A^T)$ , **null spaces**  $N(A)$  and  $N(A^T)$ . In general,  $C(A)$  is a subspace of  $\mathbb{R}^m$  and  $C(A^T)$  is a subspace of  $\mathbb{R}^n$ , both have dimension  $r = \text{rank}(A)$ . The null space of  $A$  is the set of the solutions of  $Ax = 0$ :

$$Ax = \begin{bmatrix} - & r_1^T & - \\ & \vdots & \\ - & r_m^T & - \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = 0 \quad \Leftrightarrow \quad r_i^T x = 0 \text{ for } i = 1, \dots, m$$

Recall that vectors  $u$  and  $v$  are **orthogonal** (or perpendicular) if  $u \cdot v = u^T v = 0$ , so  $N(A)$  is a subspace of  $\mathbb{R}^n$  **orthogonal** to the row space  $C(A^T)$ . The dimension of  $N(A)$ , called the **nullity** of  $A$ , is  $n - r$  by the **Rank-Nullity Theorem**: nullity + rank =  $n$ . Similarly,  $N(A^T)$  is a subspace of  $\mathbb{R}^m$  orthogonal to the row space of  $A^T$  or the column space of  $A$ ,  $C(A)$ , and the dimension of  $N(A^T)$  is  $m - r$ .

